

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center G , and points M and N lie on the circle. Line segments MH and NH are tangent to the circle at points M and N , respectively. If the radius of the circle is 168 millimeters and the perimeter of quadrilateral $GMHN$ is 3,856 millimeters, what is the distance, in millimeters, between points G and H ?

Answer

- A. 168
- B. 1,752
- C. 1,760
- D. 1,768

Correct Answer: D

Rationale

Choice D is correct. It's given that the radius of the circle is 168 millimeters. Since points M and N both lie on the circle, segments GM and GN are both radii. Therefore, segments GM and GN each have length 168 millimeters. Two segments that are tangent to a circle and have a common exterior endpoint have equal length. Therefore, segment MH and segment NH have equal length. Let x represent the length of segment MH . Then x also represents the length of segment NH . It's given that the perimeter of quadrilateral $GMHN$ is 3,856 millimeters. Since the perimeter of a quadrilateral is equal to the sum of the lengths of the sides of the quadrilateral, $3,856 = 168 + 168 + x + x$, or $3,856 = 336 + 2x$. Subtracting 336 from both sides of this equation yields $3,520 = 2x$, and dividing both sides of this equation by 2 yields $1,760 = x$. Therefore, the length of segment MH is 1,760 millimeters. A line segment that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. Therefore, segment GM is perpendicular to segment MH . Since perpendicular segments form right angles, angle GMH is a right angle. Therefore, triangle GMH is a right triangle with legs of length 1,760 millimeters and 168 millimeters, and hypotenuse GH . By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting 1,760 for a and 168 for b in this equation yields $1,760^2 + 168^2 = c^2$, or $3,125,824 = c^2$. Taking the square root of both sides of this equation yields $\pm 1,768 = c$. Since c represents a length, which must be positive, the value of c is 1,768. Therefore, the length of segment GH is 1,768 millimeters, so the distance between points G and H is 1,768 millimeters.

Choice A is incorrect. This is the distance between points G and M and between points G and N , not the distance between points G and H .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the distance between points M and H and between points N and H , not the distance between points G and H .